

Qingquan “Raphael” Xie

✉ qingqx@uw.edu ☎ 206-778-7689 📍 Seattle, WA

Education

University of Washington, Seattle

M.S. in Statistics (Expected Mar 2027)

Sept 2025 – Present

- **Current GPA:** 3.8/4.0

University of Washington, Seattle

B.S. in Applied and Computational Mathematical Sciences & B.A. in Sociology

Minor: Data Science | Cum Laude

Oct 2019 – Jun 2023

- **GPA:** 3.91/4.0
- **Annual Dean's List:** University of Washington, 2020, 2022, 2023

Skills

Programming & Data: Python, R, SQL

Statistical Methods: Causal inference, structural equation modeling, network modeling

Statistical Computing: C for performance-critical code, parallel computing, Monte Carlo simulation

Machine Learning & NLP: Tree ensembles, topic models (STM, LDA), text embeddings (GloVe, BERT), LLM agent pipelines

Publications

[1] **Xie, Q.**, Liu, Y., Zhu, Z., Xiao, K., & Li, D. (2026). Depth-dependent mechanisms underpinning soil organic carbon accumulation following afforestation. *Journal of Plant Ecology*.

[2] Zhu, Z., Xiao, K., Liu, Y., **Xie, Q.**, & Li, D. (2026). Microbial necromass dominates particulate and mineral-associated organic carbon accumulation in calcareous soil following afforestation. *Journal of Applied Ecology*.

[3] Zhu, Z., Chen, S., **Xie, Q.**, Wen, Li., Wang, K., & Li, D. (2026). Twelve-year high manure input enhances soil organic carbon stability and ecosystem multifunctionality in a karst agroecosystem. *Plant and Soil*.

[4] Fang, Y., He, X., **Xie, Q.**, & Li, D. (2025). Increasing tree phylogenetic diversity stimulates microbial functional potential in a subtropical forest. *Journal of Plant Ecology*.

[5] Liu, Q., Xie, Z., Tang, S., He, X., **Xie, Q.**, & Li, D. (2024). Synthetic microbial community enhances lignocellulose degradation during composting by assembling fungal communities. *Bioresource Technology*.

Conference

Xie, Q., & Hock, E. “Making Expert Taste Computable: Symbolic Boundaries and the Discursive Production of Value in Whisky Reviews.” Lightning talk, New Directions in Analyzing Text as Data (TADA), University of California, Berkeley, October 2026.

Research Experience

University of Washington

Research Assistant | Advisor: Prof. Tyler H. McCormick

Jan 2026 – Present

Seattle, WA

- Extended the open-source *Generative Agents* codebase, adding a perception-survey instrument ($n = 15$) covering pairwise ties, centrality, brokerage and self-position.
- Scored perceived structure against ground truth computed from the observed interaction graph.
- Logged retrieval diagnostics non-invasively by snapshotting and restoring memory-access timestamps.
- Implemented hub vs. broker network shocks (degree vs. betweenness) to test pre/post-shock perception updates.

Institute of Subtropical Agriculture, Chinese Academy of Sciences

Research Assistant | Advisor: Prof. Dejun Li

Jun 2024 – Aug 2025

China

- Identified depth-dependent drivers of soil carbon accumulation using Boruta-enhanced random forest and SEM.
- Independently built the project's end-to-end analysis pipeline in R, from preprocessing to modeling and visualization.
- Stabilized random forest feature selection by adding Boruta filtering.
- **First-author** on an accepted paper in *Journal of Plant Ecology*.

Hunan University, China

Nov 2023 – Apr 2024

Research Assistant | Advisor: Prof. Jianhua Wang

- Applied structural topic models to patent data from 139 Chinese cities to map innovation dynamics.
- Constructed randomized bipartite patent networks via MCMC to benchmark novelty.
- Applied network modeling and probabilistic inference to computational social science data.

Industry Experience

Wondershare Technology

Nov 2023 – Apr 2024

Data Analyst (Part-Time)

Changsha, China

- Analyzed **100K+** user behavior records in SQL and GA4 to support product decisions.
- Raised conversion rate **17%** in one week by tracing a UI bottleneck through funnel and churn analyses.
- Standardized event tracking across seven product lines with PMs and engineers.

Selected Graduate Course Projects

Feature selection benchmark (*STAT 534: Statistical Computing*): Reimplemented Boruta feature selection in R and benchmarked it against lasso, random forest and Metropolis–Hastings selection on accuracy and run time.

Household adult incarceration and later income (*STAT 566: Causal Modeling*): Estimated the effect of adolescent exposure to household incarceration on later family income in NLSY97, comparing outcome regression with inverse probability weighting.

Making expert taste computable (*CSSS 594: Text as Data*): Parsed 11,149 expert whisky reviews by section; descriptive sections carry 15× the score information of the closing comments (adjusted R^2 0.335 vs. 0.022); accepted as a TADA 2026 lightning talk.

Campus Involvement

Grader of STAT 220, Autumn 2026

Oct 2026 – Dec 2026

Grader of STAT 394, Spring 2026

Apr 2026 – Jun 2026

Academic Tutor, Statistics Study Center

Sept 2025 – Present

Other Projects

Freelance Whisky Writer

Jan 2026 – Present

AngeShare, China

Write blog posts about the Greater Seattle Area whisky distilleries, combining distillery history, Seattle’s urban context, in-depth whisky making process analysis, and tasting notes.